

30)
Mathematical Calculation for 2 plays

The initial values,

$$N(A) = N(B) = N(C) = 0$$

$$\theta_1(A) = \theta_1(B) = \theta_1(C) = 0$$

4

$$\epsilon = 0.2$$

formula:-

$$\theta_{\text{new}}(a) = \theta_{\text{old}}(a) + \frac{R - \theta_{\text{old}}(a)}{N(a)}$$

Play 1:

Suppose Machine A is selected

$$R = 3$$

$$N(A) = 0 + 1 = 1$$

$$\therefore \theta_1(A) = 0 + \frac{3-0}{1}$$

$$\theta_1(A) = 3$$

$$N(A) = 1, \theta_1(A) = 3$$

$$N(B) = 0, \theta_1(B) = 0$$

$$N(C) = 0, \theta_1(C) = 0$$

Play 2

$$R = 6$$

$$n(B) = 0 + 1 = 1$$

$$Q(B) = 0 + \frac{6-0}{1}$$

$$Q(B) = 6$$

Machine	$n(a)$	Reward	$Q(a)$
A	1	3	3
B	1	6	6
C	0	-	-

$$R_{\text{total}} = 3 + 6 \\ = 9 //$$

Best Machine

$$\max(Q(A), Q(B), Q(C)) \\ = \max(3, 6, 0)$$

$$\text{Best Machine} = B$$

After 2 plays, the estimated values are:-

$$Q(A) = 3, \quad Q(B) = 6, \quad Q(C) = 0$$

Total Reward is

$$[9] //$$